OPERATION

As the energized rotor begins to rotate within the generator, the spinning magnetic field induces a current into the windings of the stator coil. Once the generator begins producing sufficient current, it also provides the current needed to energize the rotor.

The Y type stator winding connections deliver the induced AC current to 3 positive and 3 negative diodes for rectification. From the diodes, rectified DC current is delivered to the vehicle electrical system through the generator battery terminal.

Although the generators appear the same externally, different generators with different output ratings are used on this vehicle. Be certain that the replacement generator has the same output rating and part number as the original unit. Refer to Generator Ratings in the Specifications section at the back of this group for amperage ratings and part numbers.

Noise emitting from the generator may be caused by: worn, loose or defective bearings; a loose or defective drive pulley; incorrect, worn, damaged or misadjusted fan drive belt; loose mounting bolts; a misaligned drive pulley or a defective stator or diode.

REMOVAL

WARNING: DISCONNECT NEGATIVE CABLE FROM BATTERY BEFORE REMOVING BATTERY OUTPUT WIRE (B+ WIRE) FROM GENERATOR. FAILURE TO DO SO CAN RESULT IN INJURY OR DAMAGE TO ELECTRICAL SYSTEM.

(1) Disconnect negative battery cable at battery.

(2) Remove generator drive belt. Refer to 7, Cooling System for procedure.

(3) Remove generator pivot and mounting bolts/nut (Fig. 2), or (Fig. 3). Position generator for access to wire connectors.

(4) If equipped, unsnap plastic cover from B+ terminal.

(5) Remove B+ cable output terminal mounting nut at rear of generator (Fig. 4), (Fig. 5), or (Fig. 6). Disconnect terminal from generator.

(6) Disconnect field wire connector at rear of generator by pushing on connector tab.

(7) Remove generator from vehicle.

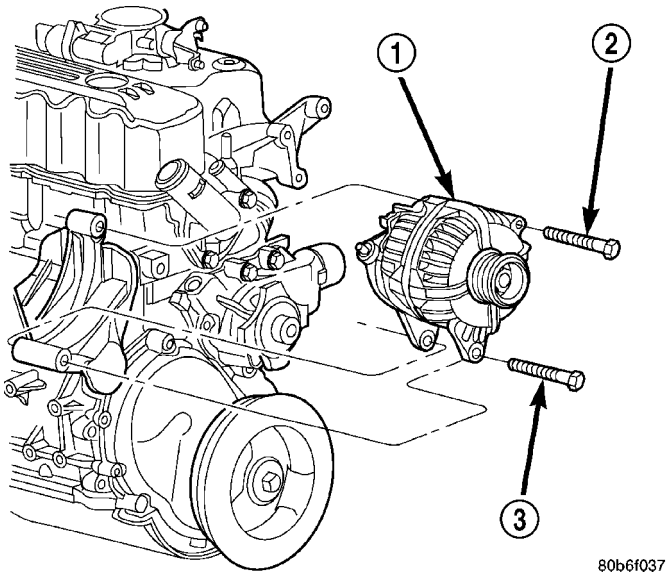

Fig. 2 GENERATOR REMOVE/INSTALL - 4.0L

1 - GENERATOR
2 - UPPER BOLT
3 - LOWER BOLT

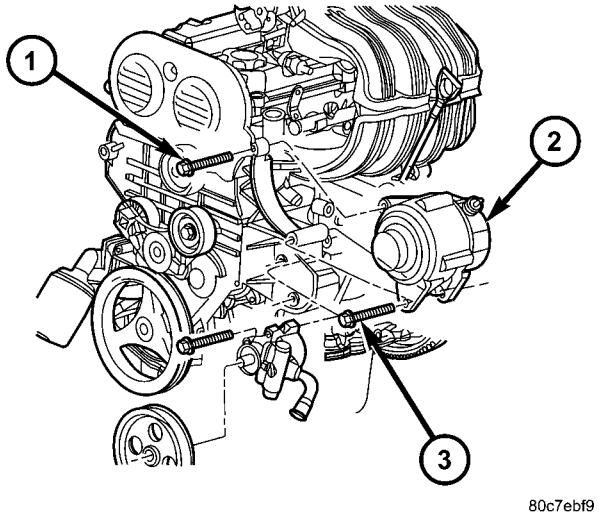

Fig. 3 GENERATOR MOUNTING- 2.4L

1 - UPPER MOUNTING BOLT
2 - GENERTOR
3 - LOWER MOUNTING BOLT